SM-P619/P613_REV0.2A

2022.04.01

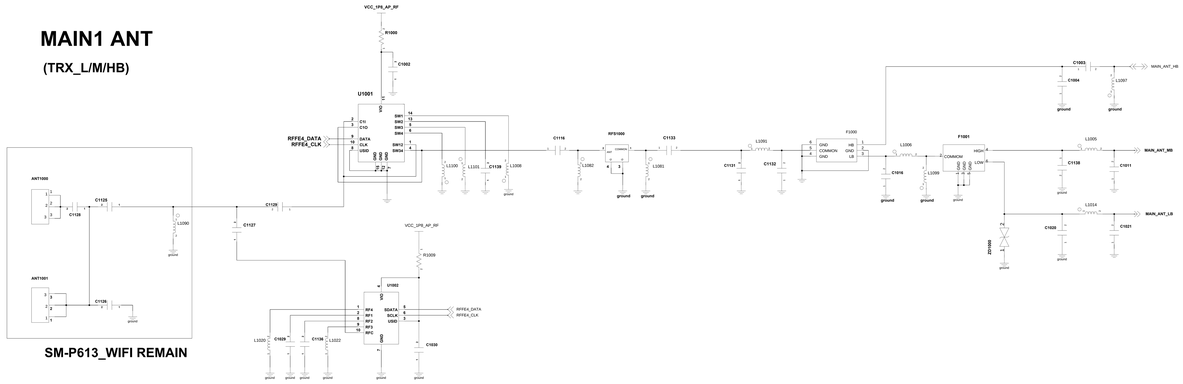
	REFERENCE
SHEET01 : MAIN RF FRONT END	1000
SHEET02 : SUB RF FRONT END, TRCV, APT	2000
SHEET03 : CONNECTIVITY, SENSORS	3000
SHEET04 : AP(SM7125)	4000
SHEET05 : AP(SM7125), eMCP	5000
SHEET06 : PMIC(Main,IF), KEY, BATTERY	6000
SHEET07 : AUDIO, SIM/SD, SHIELD CAN, IF CON	7000
SHEET08 : DISPLAY, WACOM, CAMERA	8000
SHEET09 : IF PCB, SUB ANT	9000
SHEET10 : WIFI PCB	10000

TITLE	10_SM_A505GT_LA_4_64_REV04		
SIZE A1	DWG NO PDM NUMBER		REV Rev
DRAWN BY User Name	SHEET 1 of 8	Date Drawing Date	

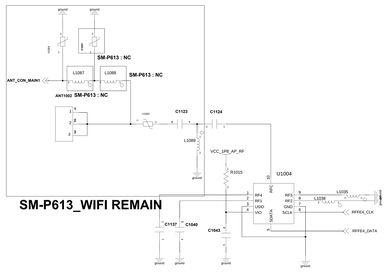
RF PART 1

MAIN1 ANT

(TRX_L/M/HB)



SM-P613_WIFI REMAIN



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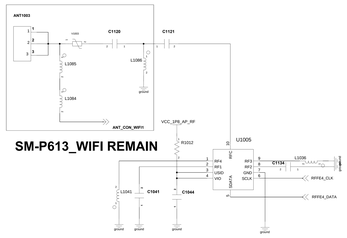

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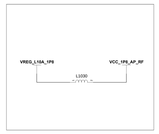
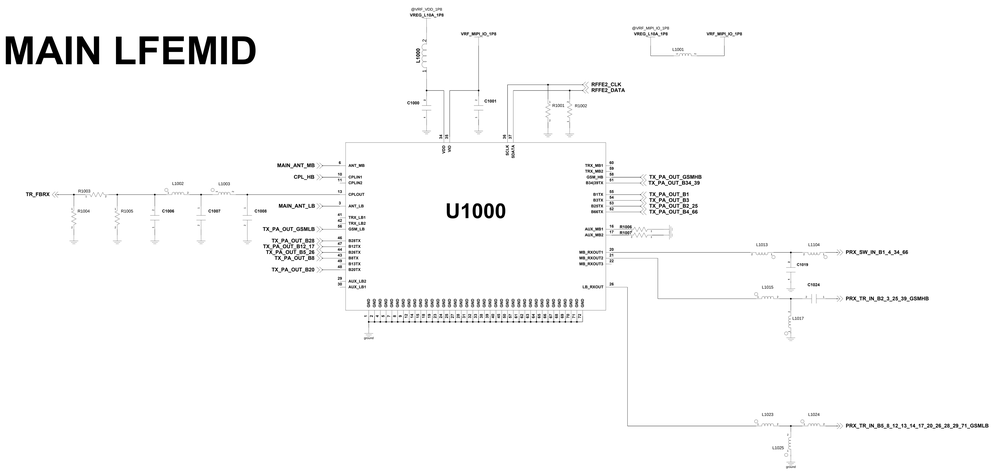


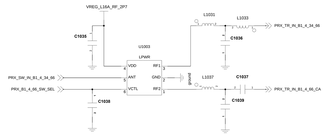
SM-P613_WIFI REMAIN

MAIN1 ANT X-GND1

MAIN1 ANT X-GND2

MAIN LFEMID





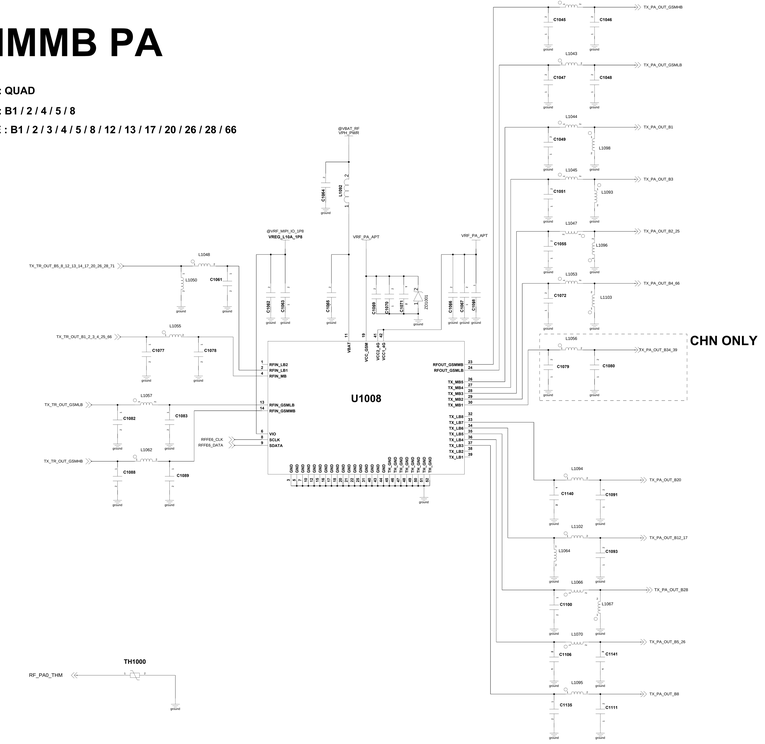
PRX B1 / B66 SW

MMMB PA

2G : QUAD

3G : B1 / 2 / 4 / 5 / 8

LTE : B1 / 2 / 3 / 4 / 5 / 8 / 12 / 13 / 17 / 20 / 26 / 28 / 66



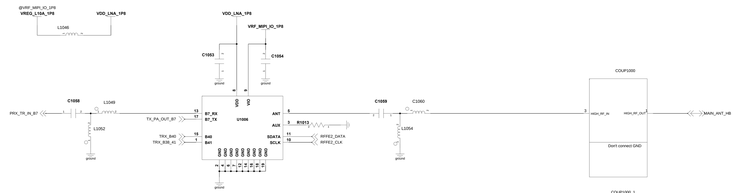
CHN ONLY

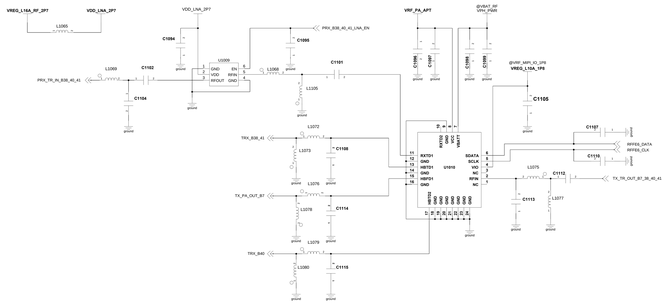
PA_THM

HIGH BAND PAMID

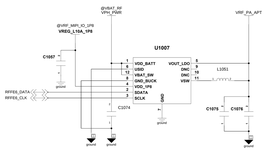
FDD LTE: B7

TDD LTE: B38 / 40 / 41





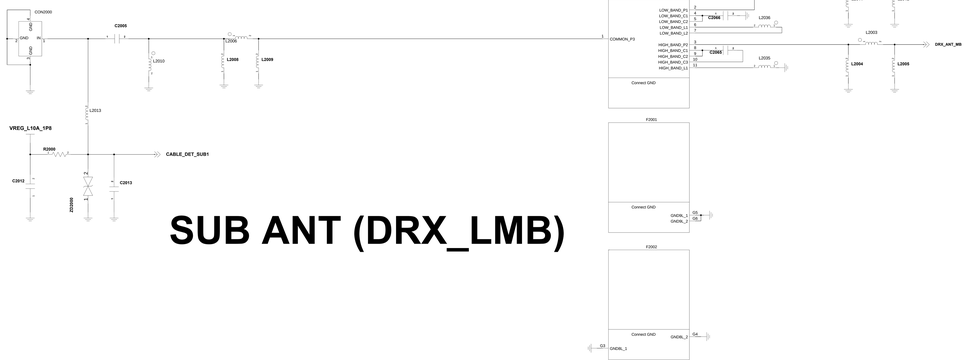
APT MODULATOR

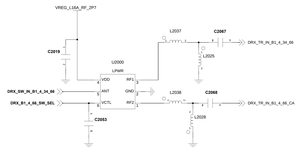


Title	10_SM_A505GT_LA_4_64_REV04		
Size A1	Number PDM NUMBER		Rev
Author User Name	Sheets 11	Date Drawing Date	

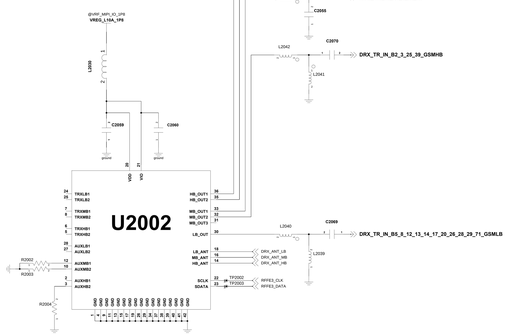
DO NOT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

SUB RF





DRX B1/B66 CA SW



DRX LFEM

SM-P613_WIFI DELETE

SUS2000 SUS2001

SCREW SUS

CLP2000 CLP2001

SM-P613_WIFI DELETE

Cable CLIP

SC2000

TCXO SHIELDCAN

GAS2000 GAS2001

Front CONTACT

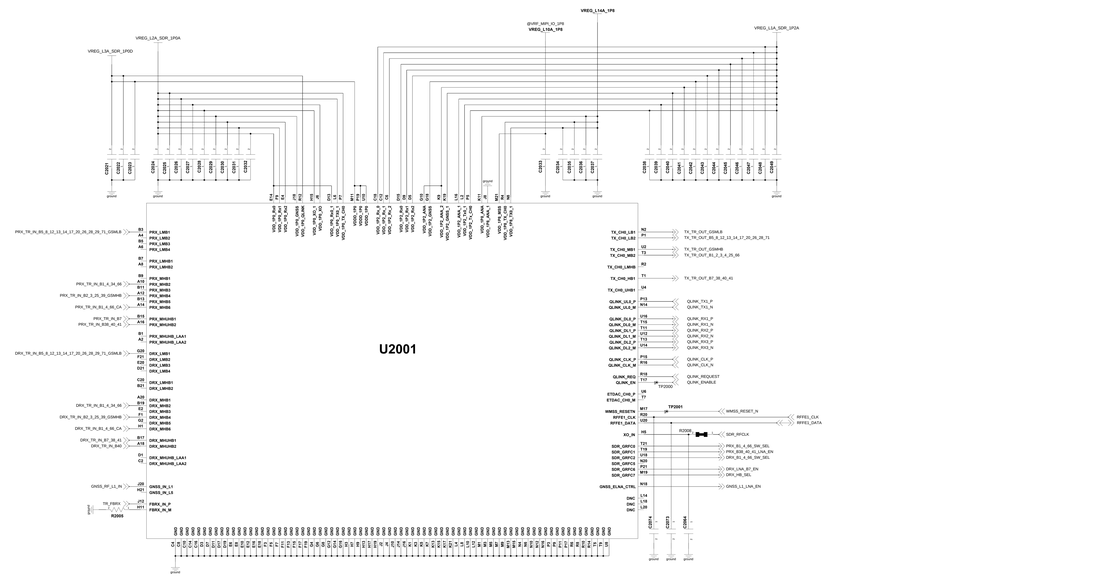
ANT2001 ANT2002 ANT2003 ANT2004 ANT2008

ANT2006 ANT2007 ANT2008 ANT2009 ANT2010 ANT2005

Rear CONTACT

ANT2011 ANT2012

SD/SIM FPCB CONTACT

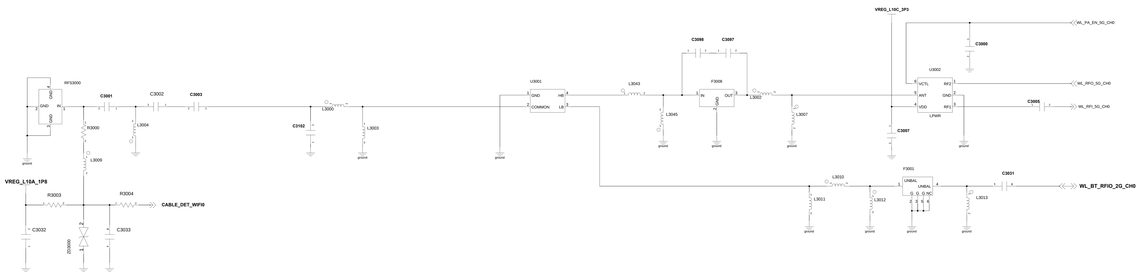


SDR675 RF TRANCEIVER

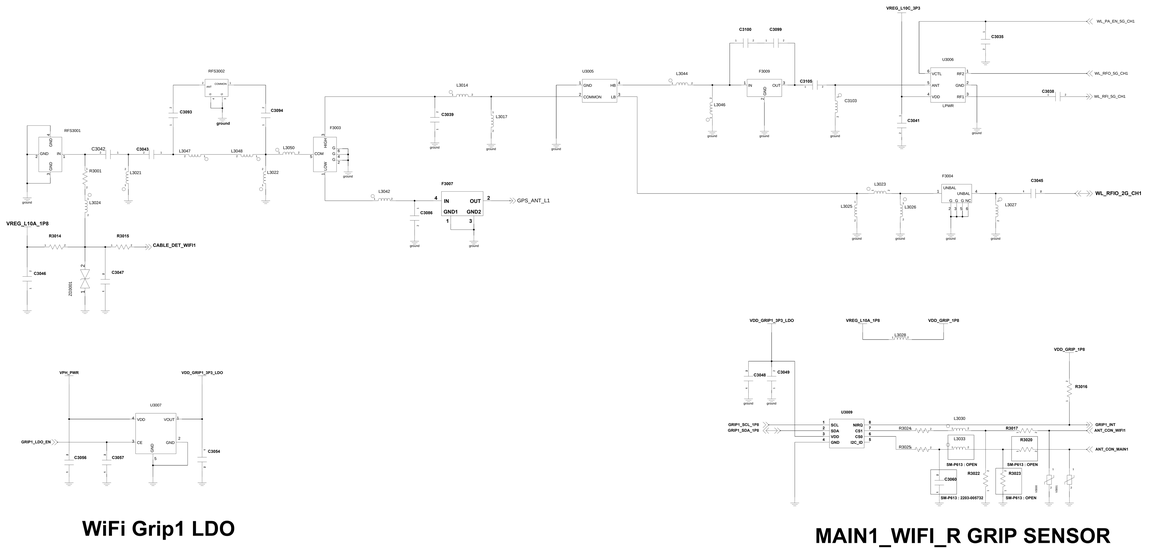
Project	10_SM_A505GT_LA_4_64_REV04		
Size	Doc No.	PDM NUMBER	Rev.
A1			
Date/Time	User Name	Drawing Date	Sheet 3 of 8

CONNECTIVITY

WIFI_0 (Right BT / WiFi 0ch)

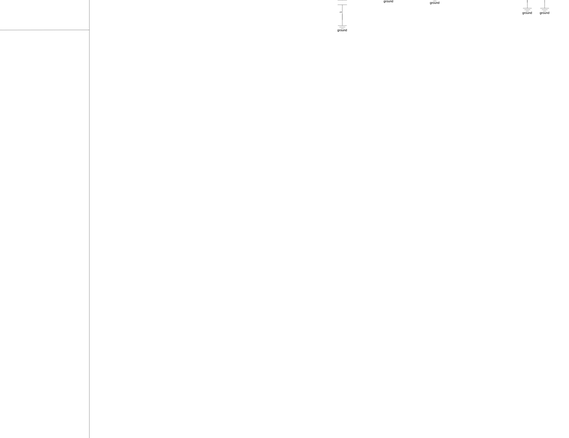


WIFI_1 (Left GNSS / WIFI 1ch)



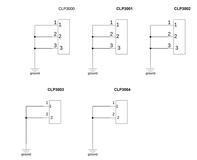
WiFi Grip1 LDO

MAIN1_WIFI_R GRIP SENSOR



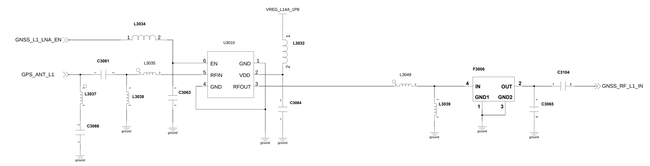
TH3000

WIFI_THM

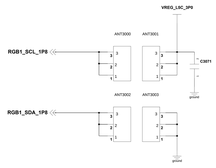


Cable CLIP

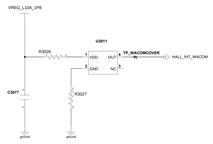
GPS L1



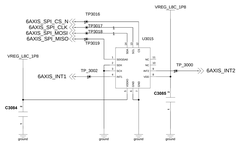
SENSOR



RGB SENSOR



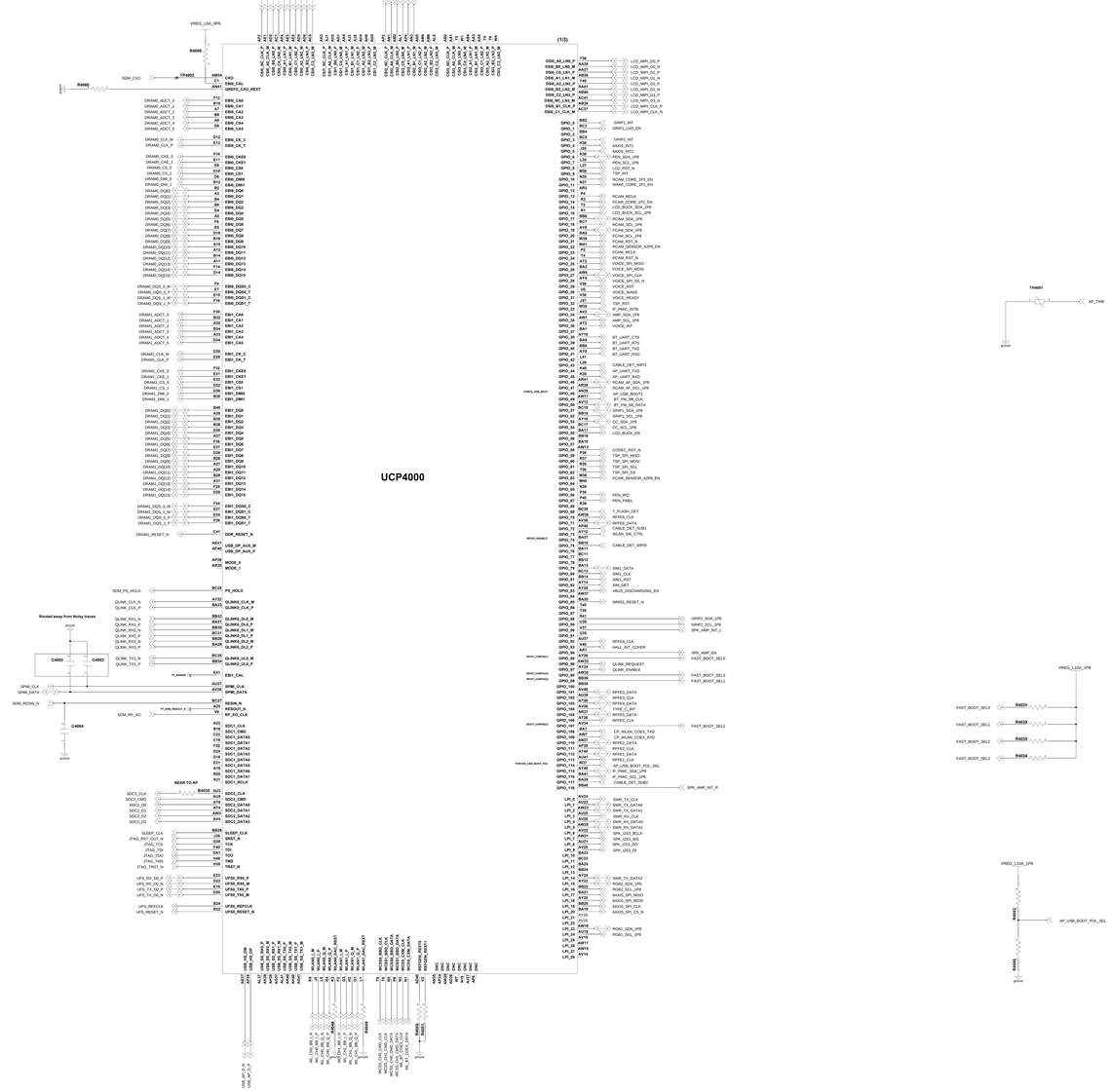
HALL_IC



6 AXIS(ACC+GRYO)

TITLE	10_SM_A505GT_LA_4_64_REV04
SIZE	A1
DOC NO.	PCM NUMBER
REV	Rev
DRAWN BY	User Name
DATE	Drawing Date
SHEET	4 of 8

AP (SM7125)


UCP4000
(1/3)
VREG_L6A_1P8
R4088
SDM_CHO
R4092
TP4002
SDM_PS_HOLD
QLINK_CLK_N
QLINK_CLK_P
Routed away from Noisy traces
QLINK_RX1_N
QLINK_RX1_P
QLINK_RX2_N
QLINK_RX2_P
QLINK_RX3_N
QLINK_RX3_P
QLINK_TX1_N
QLINK_TX1_P
C4052
C4053
SPMI_CLK
SPMI_DATA
SDM_RESIN_N
SDM_RF_HO
C4054
NEAR TO AP
R4035
R4036
R4037
R4038
R4039
R4040
R4041
R4042
R4043
R4044
R4045
R4046
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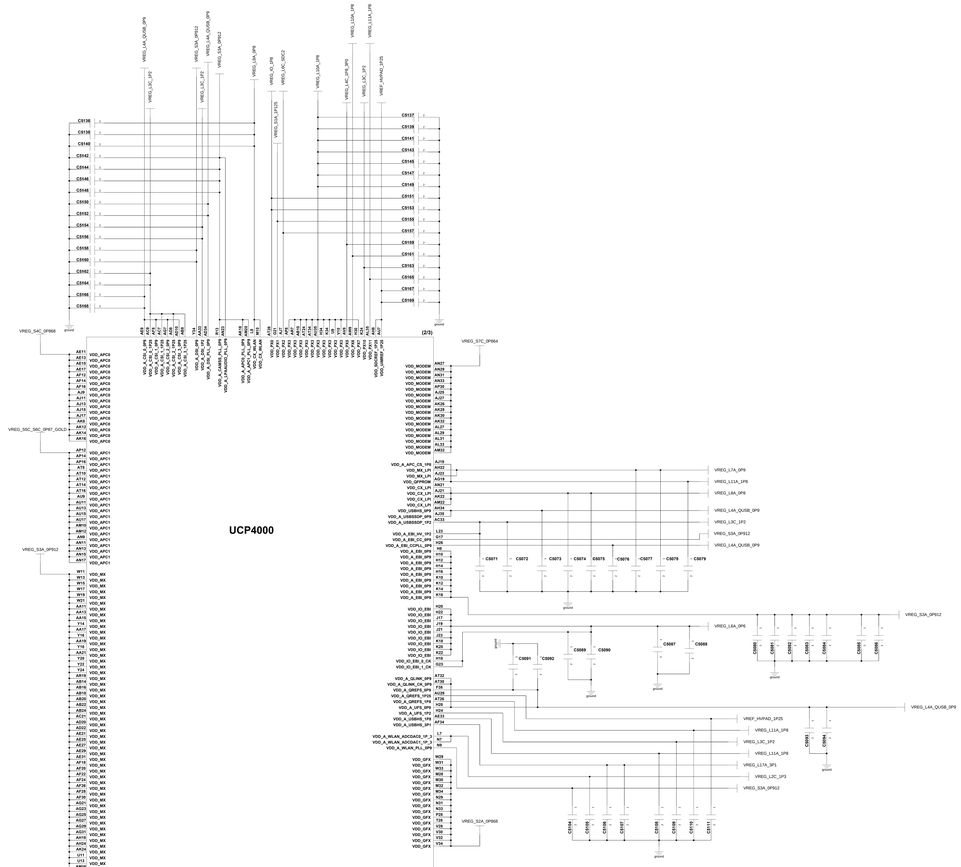
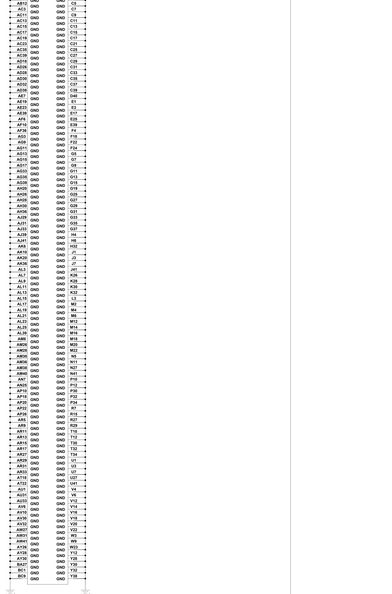
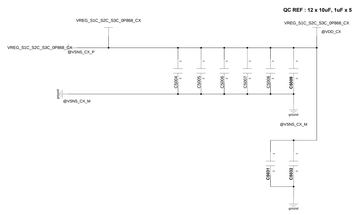
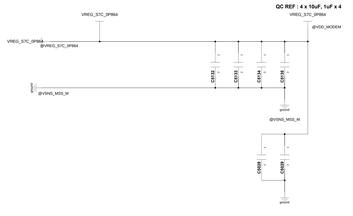
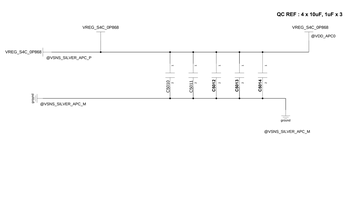
AP (SM7125, uMCP)

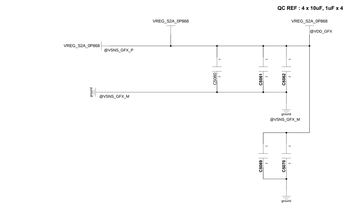


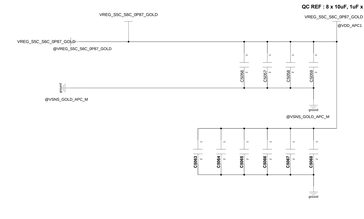


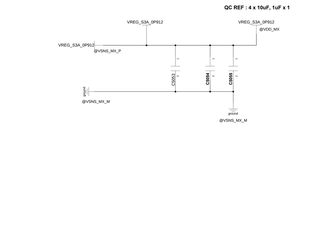


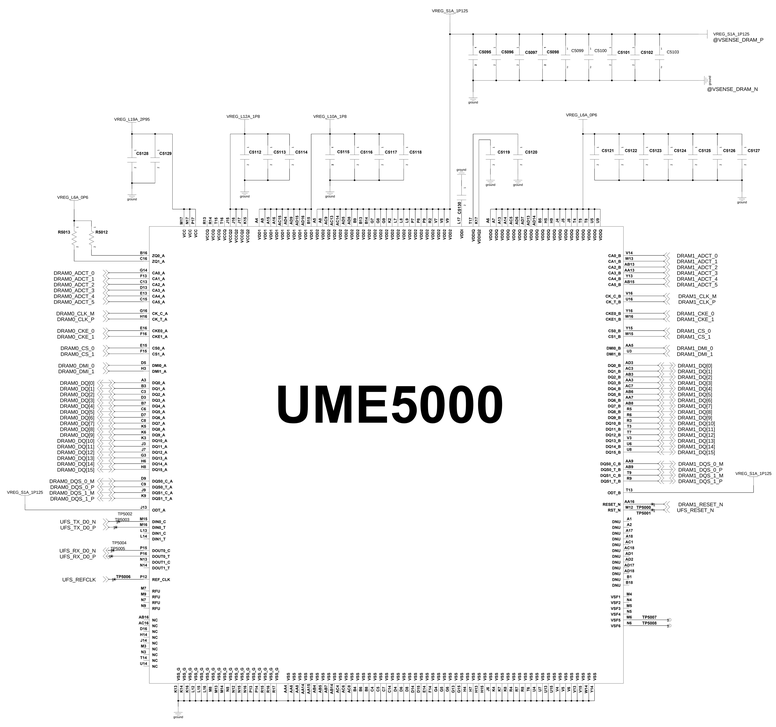












UME5000

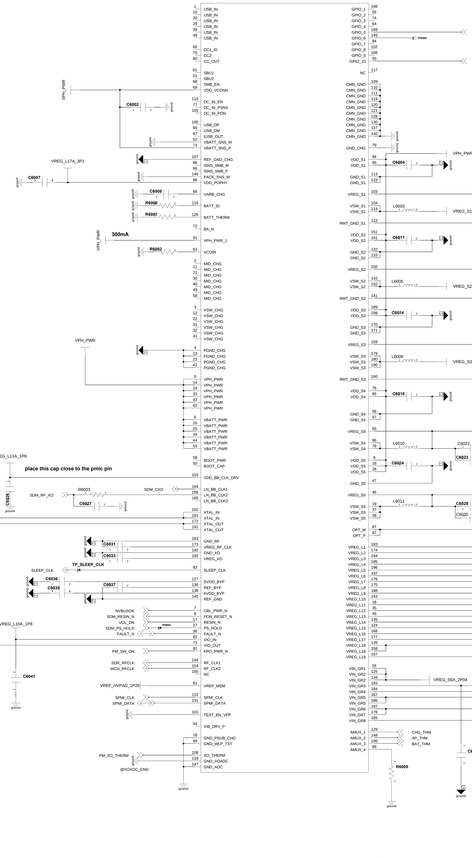
4GB + 64GB uMCP

4GB + 64GB uMCP : KM5P9001DM-B424 (1108-000720)
4GB + 128GB uMCP : KM5L9001DM-B424 (1108-000713)

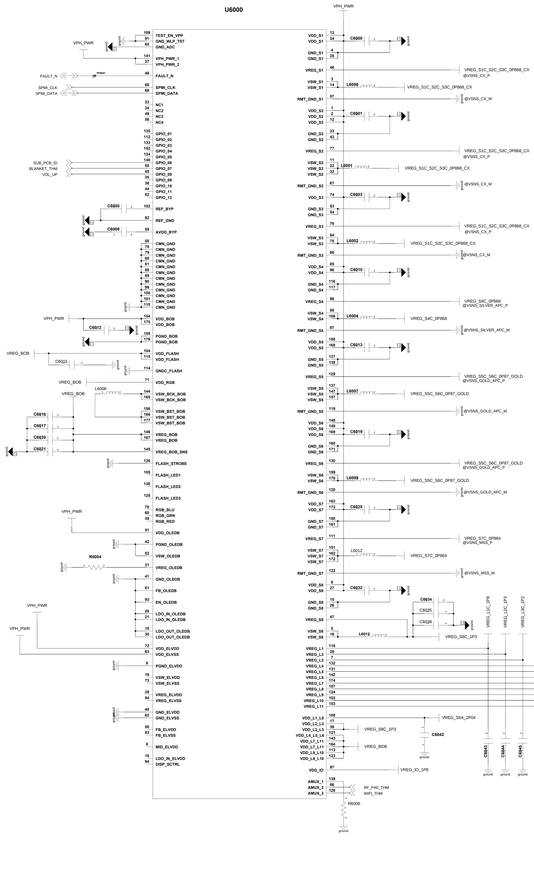
TITLE	10_SM_A505GT_LA_4_64_REV04		
A1	DOC NO	POM NUMBER	Rev
REVISION	User Name	SHEET 11	Drawing Date

POWER

U6001



U6000



PCB H/W Revision

AMUX_1				
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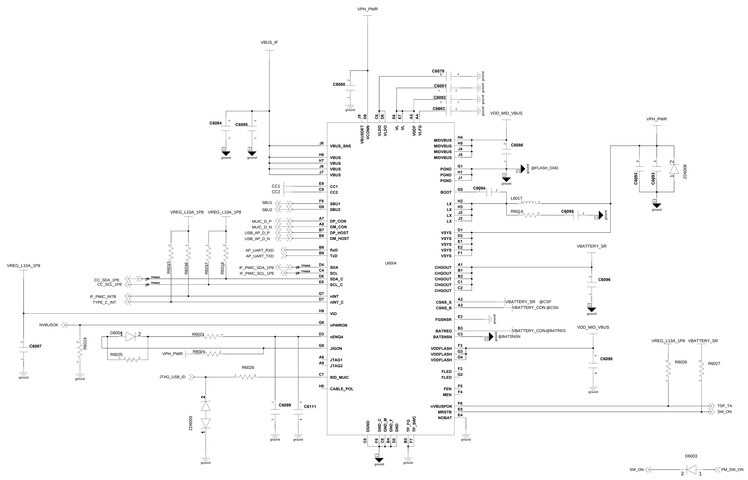
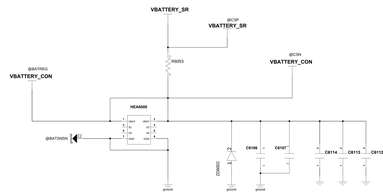
BringUp	4.7K			00
REV00	12K			01
REV01	20K			02
NOT USED	30K			03
REV02	39K			04
REV03	51K			05
NOT USED	68K			06
REV03B	82K			07
	100K			
	120K			
	150K			
	200K			

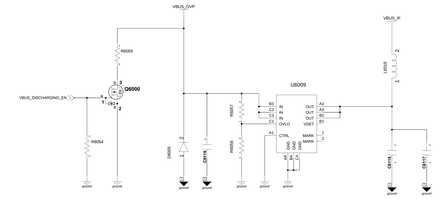
MAIN PMIC



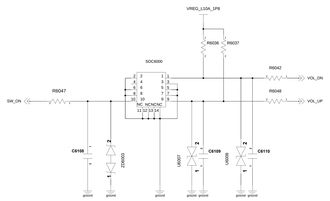
SUB PMIC



BATTERY CON



OVP



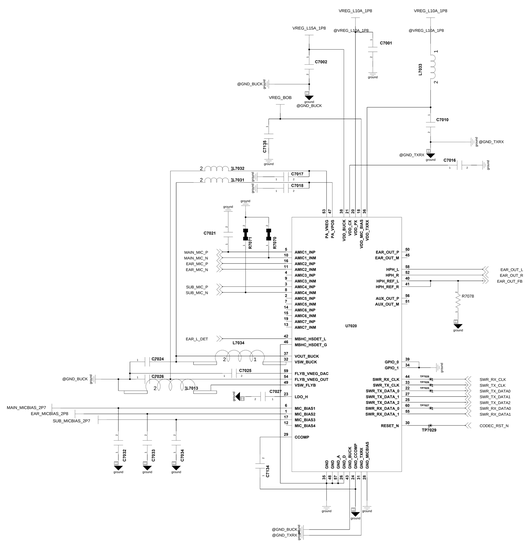
KEY CONN

CHG THM

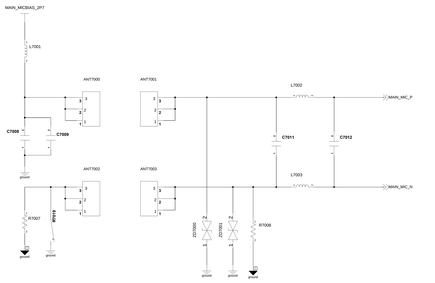
BAT THM_Option

FILE	10_SM_A505GT_LA_4_64_REV04.1.sch			
SIZE	DWG NO			REV
A1	PDM NUMBER			Rev
APPROVED	User Name	DATE	SHEET 6 OF 11	Drawing Date

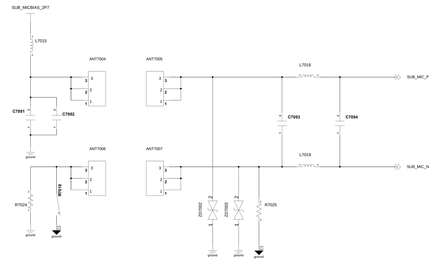
AUDIO



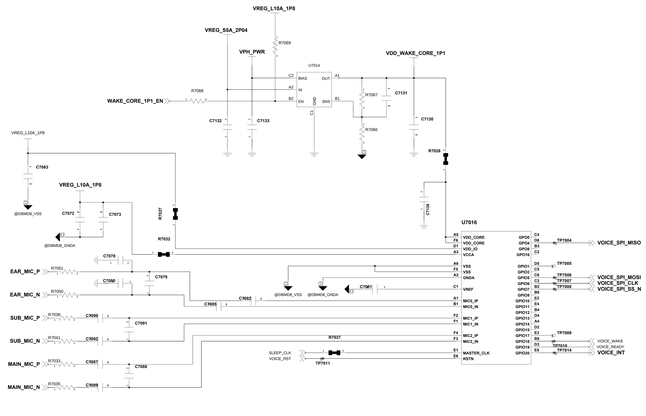
Audio CODEC



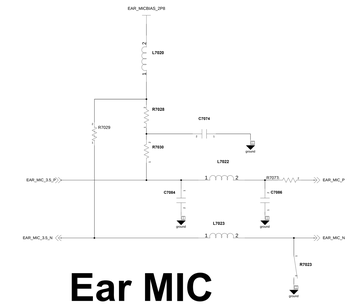
MAIN MIC



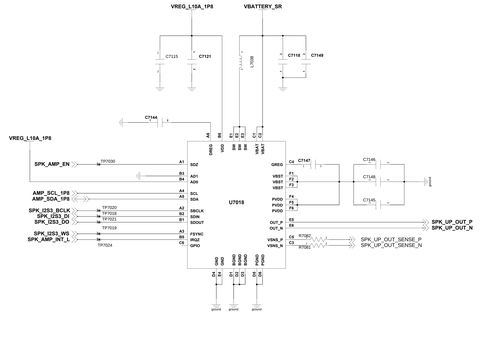
SUB MIC



WAKE UP IC



Ear MIC



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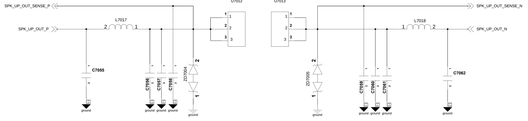
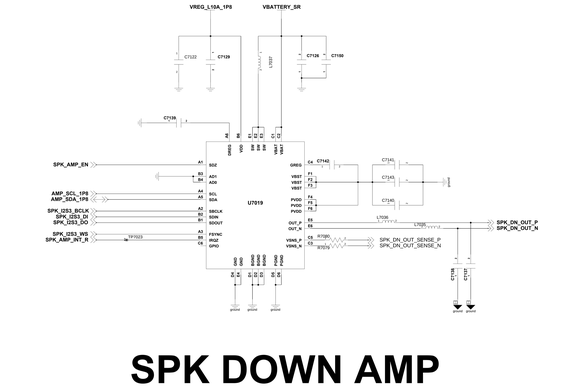
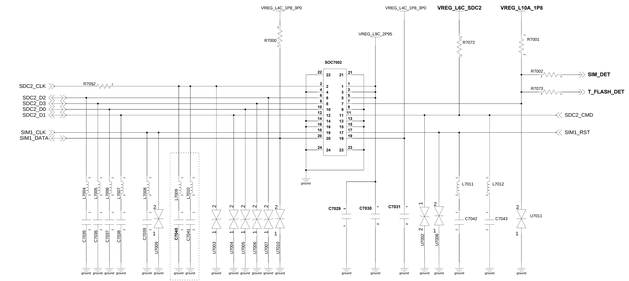


SPK UP CONTACT

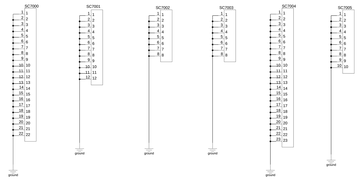


SPK DOWN AMP

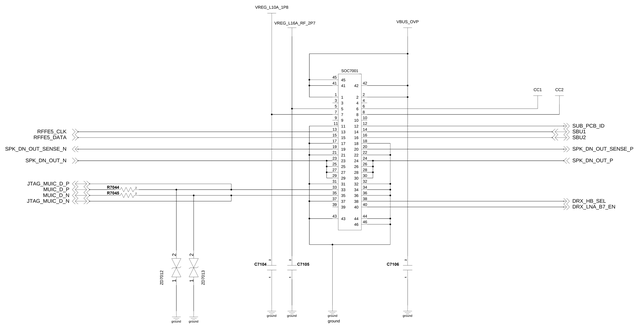
SIM & SD CON



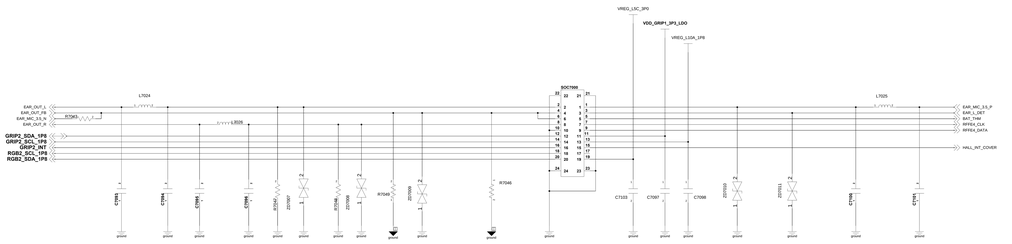
SHIELD_CAN



IF Connector

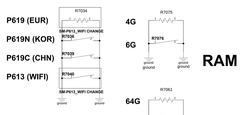


3.5pi Connector



2D BARCODE

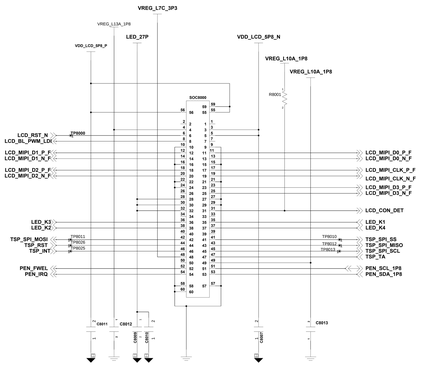
	EUR	KOR	CHN	WIFI	4G	6G	64G	128G
R7034	Solder							
R7036		Solder						
R7039			Solder					
R7040				Solder				
R7075					Solder			
R7076						Solder		
R7061							Solder	
R7062								Solder



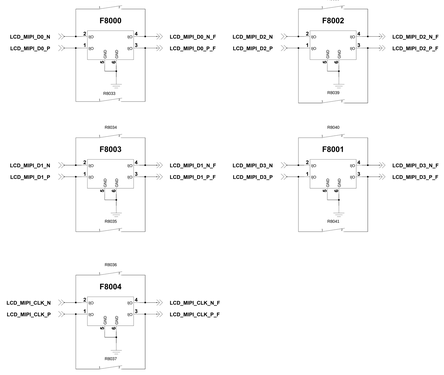
10_SM_A505GT_LA_4_64_REV04C_190405_FINAL

		PCW NUMBER	Rev
Drawn by	User Name	Size A1	Drawing Date

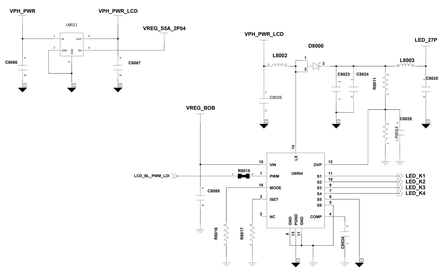
DISPLAY



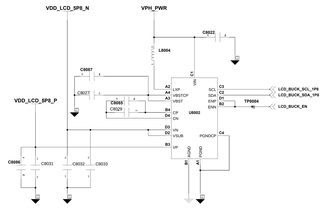
LCD CON



LCD FILTER



LCD BACKLIGHT DRIVER IC

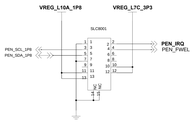


LCD BUCK BOOSTER

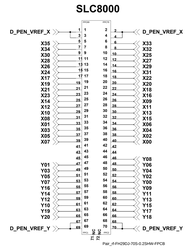
WACOM PCB

< SMD F/T >

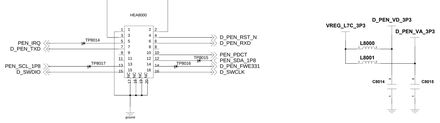
1. VREG_L7C_3P3
2. VREG_L10A_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. GND



WACOM I/F CON



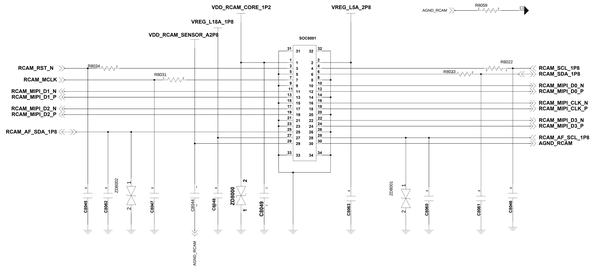
WACOM CON



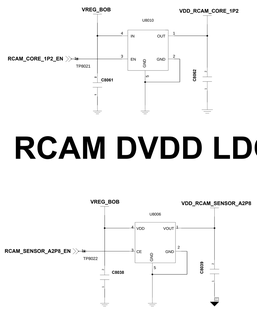
DEBUGER

S_PEN Control IC

CAMERA

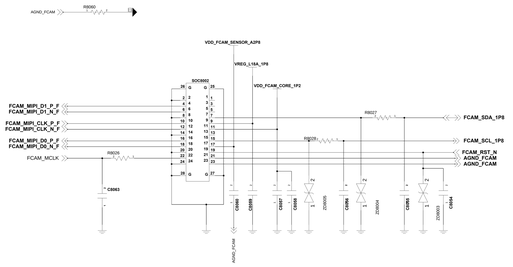


8M CAM(R_CAM)

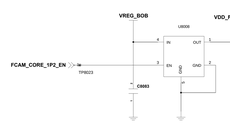


RCAM DVDD LDO

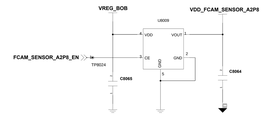
RCAM AVDD LDO



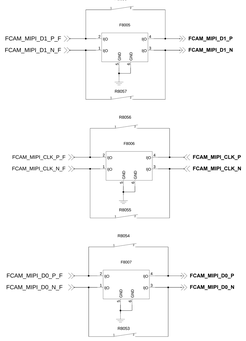
5M CAM (F_CAM)



FCAM DVDD LDO



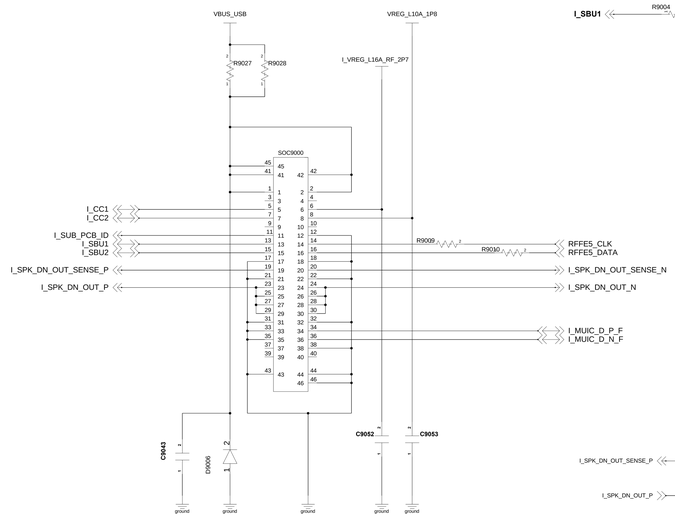
FCAM AVDD LDO



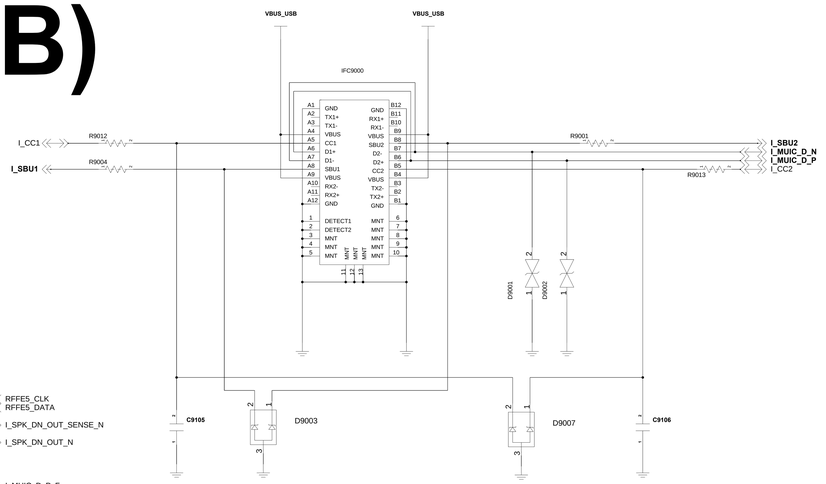
F_CAM Filter

TITLE		10_SM_A505GT_LA_4_64_REV04
APPROVED	User Name	SIZE A1 / DOCUMENT NUMBER / Rev
CHECKED		SHEET 8 of 8 / Drawing Date

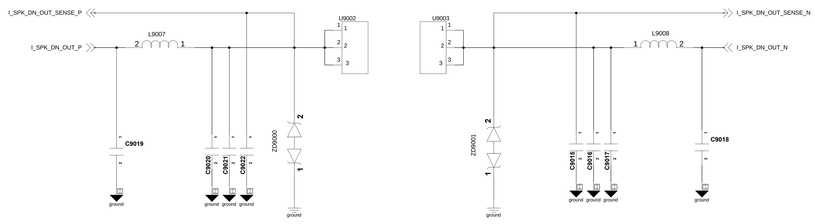
IF PCB (USB)

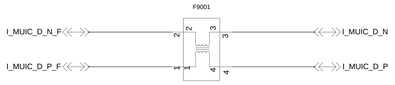


IF Con.



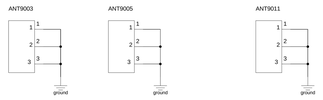
USB C-TYPE CONN





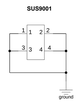
SPK DOWN CONTACT

USB 2.0 CMF

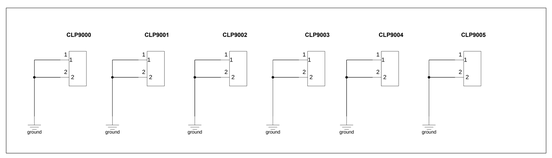


1.3T 0.9T

REAR CONTACT

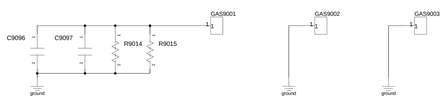


SCREW SUS(0.3T)

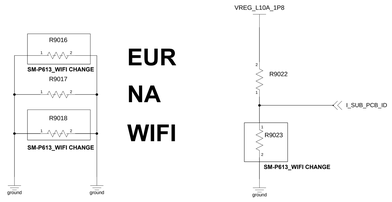


Cable CLIP

SM-P613_WIFI DELETE



Front CONTACT



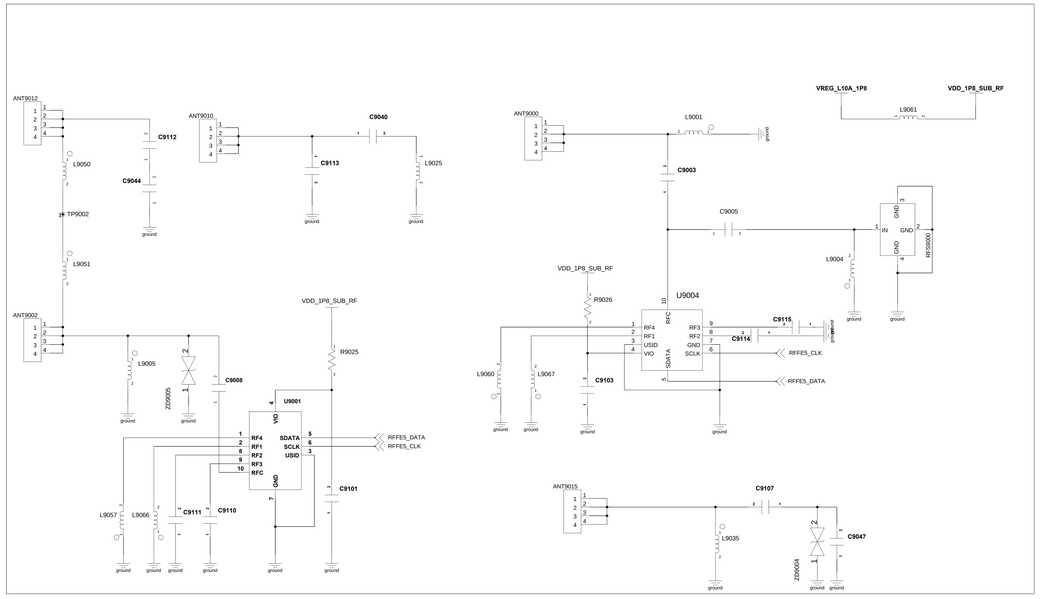
EUR

NA

WIFI

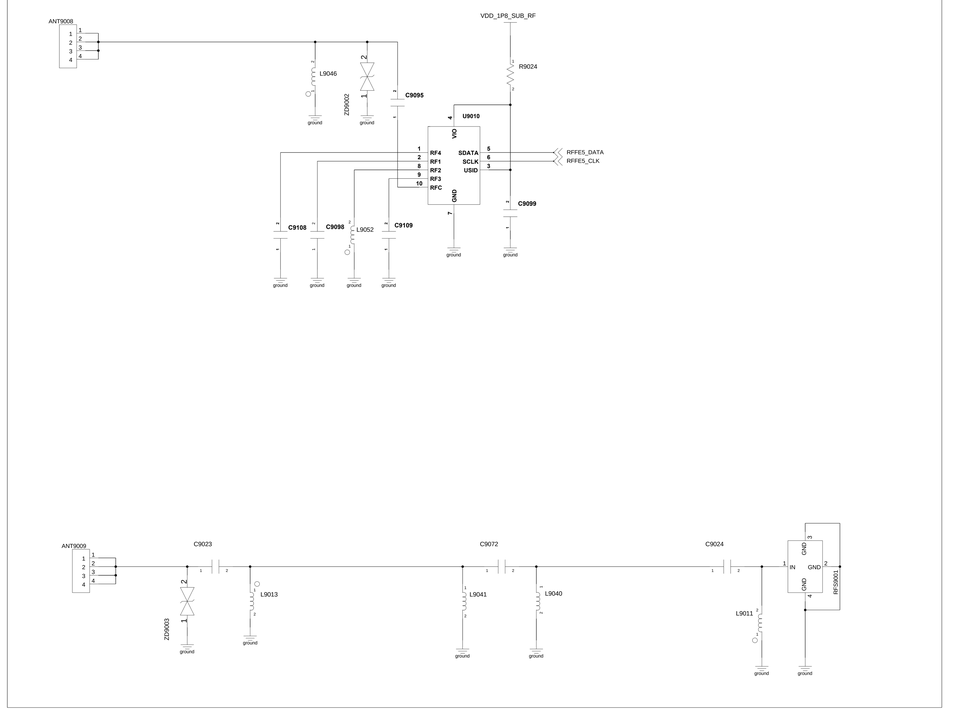
I_SUB_PCB_ID (PM6150A : GPIO_06)	Region	R9023	ADC	Range	Voltage
1	EUR		324	0 - 600	0.32V
2	NA		728	601 - 900	0.72V
3	Reserve				0.9V
4	WIFI		1080	901 - 1200	1.08V
5	ETC		1381	1201 - 1500	1.38V

SUB ANT1(LM)



SM-P613_WIFI DELETE

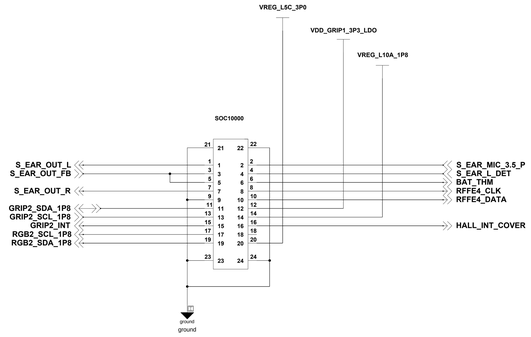
SUB ANT2(HB)



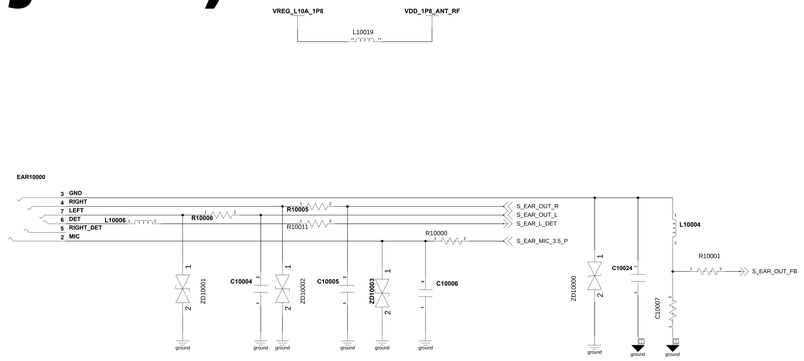
SM-P613_WIFI DELETE

TITLE	10_SM_A505GT_LA_4_64_REV04C_1					
SIZE	A1	DWG NO	PDM NUMBER		REV	Rev
DRAWN BY	User Name	SHEET	17	of	9	Date Drawing Date

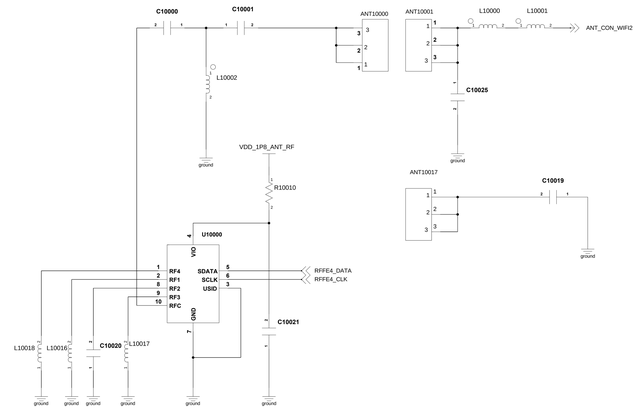
3.5pi PCB (Earjack)



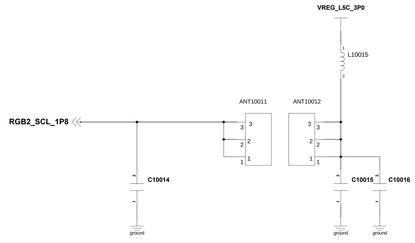
WIFI IF Con.



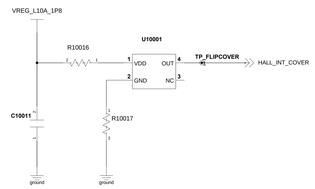
EARJACK



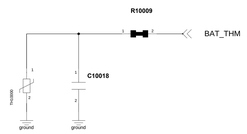
WIFI_1 ANT



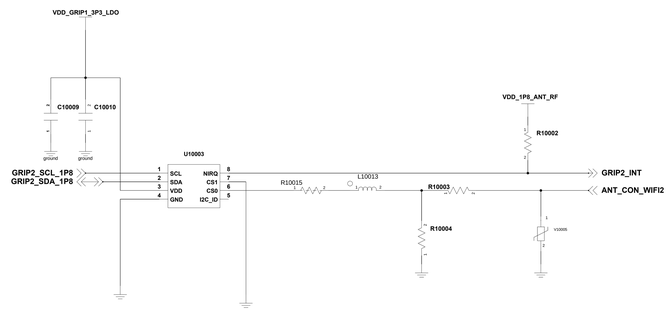
RGB SENSOR CON



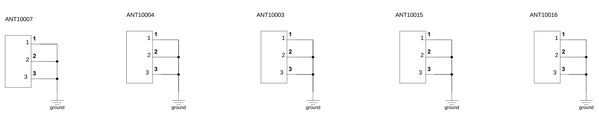
HALL_IC



BAT_THM

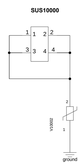


WIFI GRIP SENSOR



REAR CONTACT

Front CONTACT

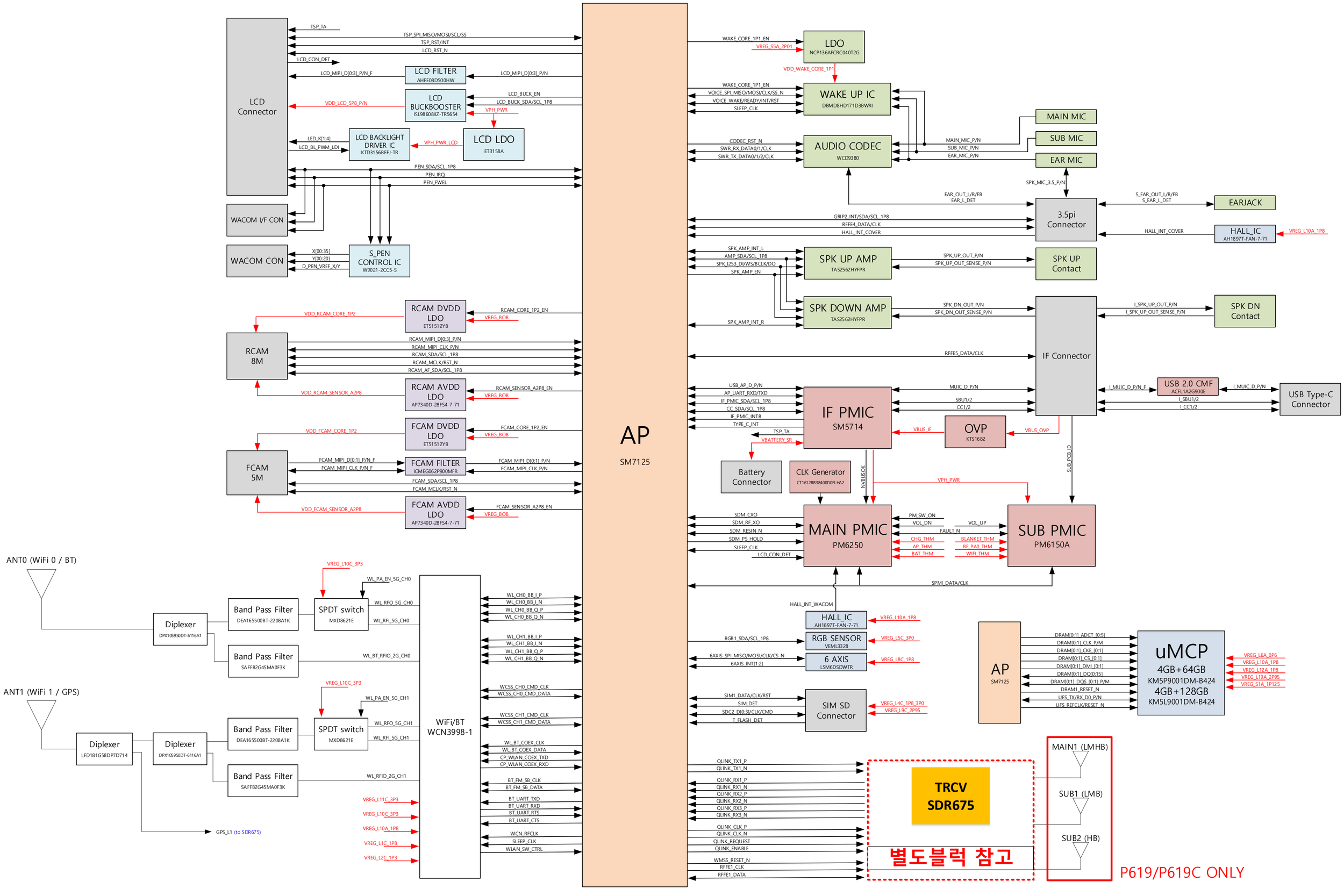


Screw SUS

TITLE	10_SM_A505GT_LA_4_64_REV04		
SIZE	A1	DOC NO	PDM NUMBER
		REV	Rev
DRAWN BY	User Name	SHEET	SHEET10 11
		DATE	Drawing Date

Tab S6 Lite SM7125 Block Diagram

SM-P619/P613 Common



Tab S6 Lite Refresh_SDR675

SM-P619/P613 Common

P619/P619C ONLY

MAIN ANT1 (L/M/HB)

2G : QUAD
WCDMA : 1/2/4/5/8
LTE FDD : 1/2/3/4/5/7/8/12/13/17/20/26/28/32/66
LTE TDD : 34/38/39/40/41

ANT_CON_MAIN1

Ant Tuner
QAT3555
(RFFE4 / USID_Low)

RF Switch
MS-156C

DIPLEXER
DPX252690DT-5075C1K

DIPLEXER
DPX202700DT-4062A1

Ant Switch
QAT3516
(RFFE4 / USID_Low)

Ant Switch
QAT3516
(RFFE4 / USID_High)

MAIN_ANT_MB

MAIN_ANT_LB

CPL_HB

TR_FBRX
Go to TRCV

MAIN LFEMiD
(RFFE2)
LMSWKSGU-K21

ANT_M
CPL_In1
CPL_In2
CPL_Out
ANT_L
Matching
TERM
VDD
VIO
SCLK
SDATA
100pF

B1Tx
B3Tx
B25Tx
B66Tx
AUX_MB1
AUX_MB2

MB_Rx out1
B1/66/34Rx
MB_Rx out2
B25/3/39Rx/AUX
MB_Rx out3
B32Rx/AUX
GHTx
B34/39Tx
TRX_MB1
TRX_MB2
LB_Rx out
B8
B26
B20
B12
B13
B28
B29Rx
AUX_LB1
AUX_LB2
B8Tx
B26Tx
B20Tx
B12Tx
B13Tx
B28Tx
GLTx
TRX_LB1
TRX_LB2

PRX_SW_IN_B1_4_66

PRX_B1_4_66_CA_SW
MXD8621E

From TRCV (GRFC0)
PRX_B1_4_66_SW_SEL

PRX_TR_IN_B1_4_34_66
PRX_TR_IN_B1_4_66_CA
PRX_TR_IN_B2_3_25_39_GSMHB
PRX_TR_IN_B5_8_12_13_14_20_26_28_29_71_GSMLB

PRX_MHB2
PRX_MHB6
PRX_MHB4
PRX_MHB1
PRX_LMB1

MMMB PA
(RFFE6)
QM58110

VRF_PA_APT

APT Modulator
(RFFE6)
QET4101

VCC_4G
LB1
LB2
LB3
LB4
LB5
LB6
LB7
LB8
MB1
MB2
MB3
MB4
MB5
LB1/5/8_IN_1
LB1/5/8_IN_2
MB2/4_IN
VIO
SCLK
SDATA
VBATT
MB2/4_OUT
LB2/5_OUT
MB2/4_IN
LB2/5_IN
VCC_2G

TX_PA_OUT_B8
TX_PA_OUT_B5_26
TX_PA_OUT_B28
TX_PA_OUT_B12_17
TX_PA_OUT_B20
TX_PA_OUT_B34_39
TX_PA_OUT_B4_66
TX_PA_OUT_B2_25
TX_PA_OUT_B3
TX_PA_OUT_B1
TX_PA_OUT_GSMHB
TX_PA_OUT_GSMLB

TX_TR_OUT_B5_8_12_13_14_17_20_26_28_71
TX_TR_OUT_B1_2_3_4_25_66
TX_TR_OUT_GSMHB
TX_TR_OUT_GSMLB

TX_CH0_LB2
TX_CH0_MB2
TX_CH0_MB1
TX_CH0_LB1

Go to SW (U1003)
Go to SW (U1009)
Go to SW (U2000)
Go to SW (U9008)
Go to SW (U9007 &U9009)

PRX_B1_4_66_SW_SEL
DRX_B38_40_41_LNA_EN
DRX_B1_4_66_SW_SEL
DRX_LNA_B7_EN
DRX_HB_SEL

SDR_GRFC0
SDR_GRFC1
SDR_GRFC2
SDR_GRFC6
SDR_GRFC7

TRANCEIVER
(RFFE1)
SDR675

GRIP SENSOR #1
ISG6320
ANT_CON_MAIN1
ANT_CON_WIFI1
GRIP1_INT (to SMT128)

GRIP SENSOR #2
ISG6320
ANT_CON_WIFI2
GRIP2_INT (to SMT128)

HB LFEM
(RFFE2)
SFMD3L0F001

Module
SW+LNA
Mipi 2.1
CPL
MAIN_ANT_HB
CPL_HB
AUX
B40 TRx
B41 TRx
B7 Tx
B7 Rx
VDD
VIO
SCLK
SDATA
TRX_B40
TRX_B38_41
TX_PA_OUT_B7

PRX_B38_40_41 LNA
MXD8921HH
PRX_B38_40_41_LNA_EN
From TRCV (GRFC1)

HB PAM
(RFFE6)
WIPS33232-01

VRF_PA_APT

APT Modulator
(RFFE6)
QET4101

TX/RX
Switch
PA
PA & Switch Control
VCC
RFIN
VBATT
VIO
SDATA
SCLK
RXTX1
HBTX1
HBRX1
HBTX2
RXTX2

TX_TR_OUT_B7_38_40_41
TX_CH0_HB1

PRX_TR_IN_B7
PRX_TR_IN_B38_40_41
PRX_MHUHB1
PRX_MHUHB2

SUB2 ANT (HB)

LTE FDD : 7
LTE TDD : 38/40/41

SUB1 ANT (L/MB)

2G : QUAD
WCDMA : 1/2/4/5/8
LTE FDD : 1/2/3/4/5/8/12/13/17/20/26/28/32/66
LTE TDD : 34/39

Ant Switch
QAT3516
(RFFE5 / USID_Low)

Ant Switch
QAT3515
(RFFE5 / USID_High)

Ant Switch
QAT3515
(RFFE5 / USID_Low)

Coaxial cable
Coaxial cable

DIPLEXER
DIPLEXER-LBMHB-10L-3ST-065T
(PCB embeded)

DRX_ANT_HB
DRX_ANT_MB
DRX_ANT_LB

DRX LFEM
(RFFE3)
SFMLHT0W001

TRXHB1
AUXHB1
HB_ANT
HB_OUT1
HB_OUT2
B40
B7
B41
MB_ANT
MB_OUT1
MB_OUT2
B1(4)
B3
B66
B25
B34
B39
B8
B20
B26
B12/B17
B28
B29
LB_ANT
LB_OUT
TRXLB1
AUXLB2
TRXLB2
AUXLB1

DRX_SW_IN_B1_4_34_66

DRX_B1_4_66_CA_SW
MXD8621E

From TRCV (GRFC2)
DRX_B1_4_66_SW_SEL

DRX_TR_IN_B40
DRX_TR_IN_B7_38_41
DRX_TR_IN_B1_4_34_66
DRX_TR_IN_B1_4_66_CA
DRX_TR_IN_B2_3_25_39_GSMHB
DRX_TR_IN_B5_8_12_13_14_17_20_26_28_29_71_GSMLB

DRX_MHUHB2
DRX_MHUHB1
DRX_MHB2
DRX_MHB6
DRX_MHB4
DRX_MHB1
DRX_LMB1

WiFi1 & GPS ANT

ANT_CON_WIFI2

DIPLEXER
LFD181GS8DP7D714

To WCN3998-1

GPS

GPS SAW
SFHG859F302

GPS LNA
MXD1N1655

GPS SAW
SFHG859F302

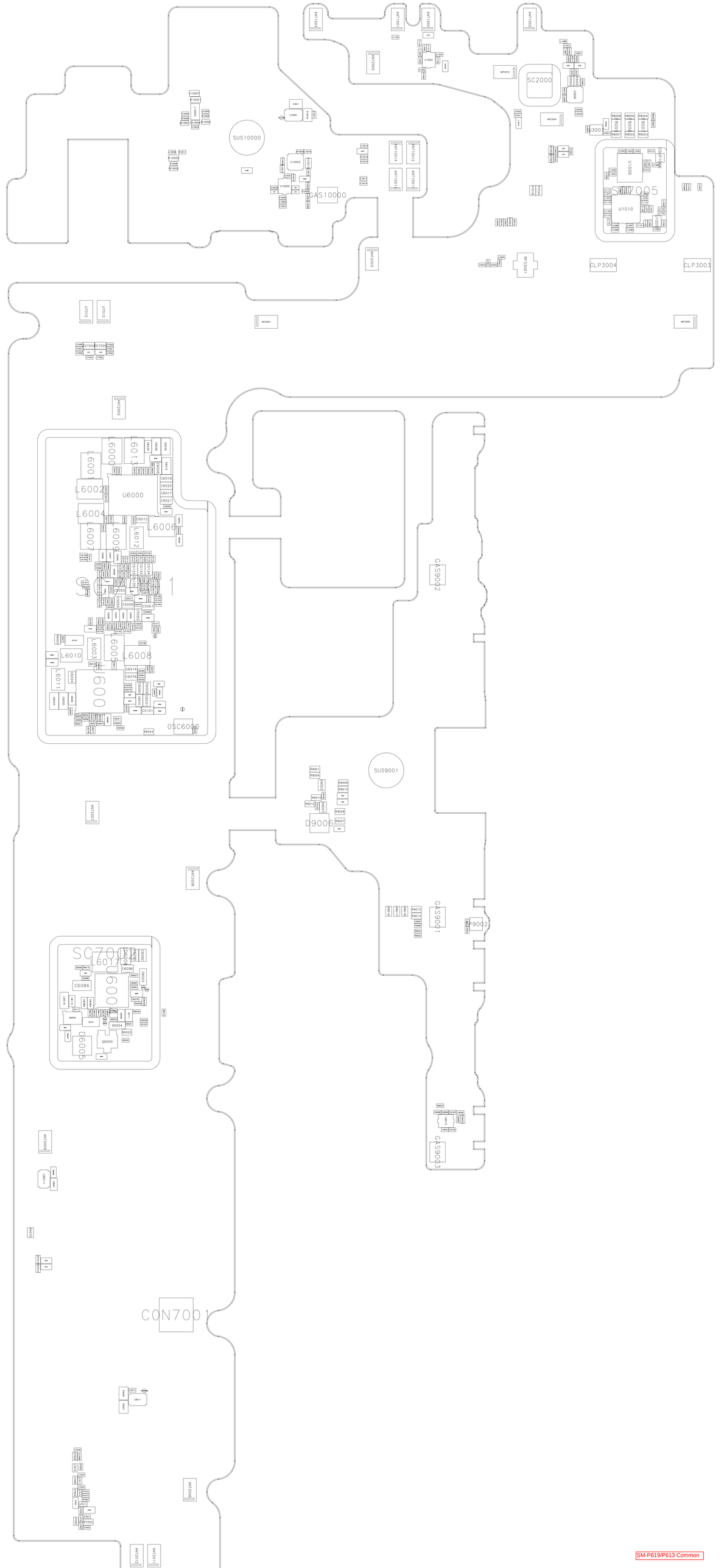
GNSS_RF_L1_IN
GNSS_IN_L1

WiFi0 & BT ANT

ANT_CON_WIFI1

To WCN3998-1

SM-P619/P613 Common


ANT1001
ANT1002
ANT2205
SC2000
ANT2004
U3001
R8054
R8056
R8024
R8055
R8057
R8053
SUS10000
U10003
GAS10000
ANT1004
ANT1012
ANT1011
U0106
SC0105
U1010
CLP1001
CLP3004
CLP3003
ANT2203
R9300
ANT2501
ANT2000
U0913
U0912
D7004
D7005
ANT2302
L6002
U6000
L6004
L6006
L6012
L6007
L6009
L6005
L6008
L6010
L6011
C6024
C6076
C6014
C6016
C6020
C6017
C6021
C6012
C5061
C5101
OSC6000
R6044
ANT2202
SUS9001
D9006
R9001
R9004
R9008
R9010
R9013
R9012
R9028
R9027
GAS9002
GAS9001
P9002
GAS9003
SC7001
C6086
C6096
C6004
D6004
R6025
D6000
R6024
ANT3000
U3011
ANT2006
ANT2301
ANT2302